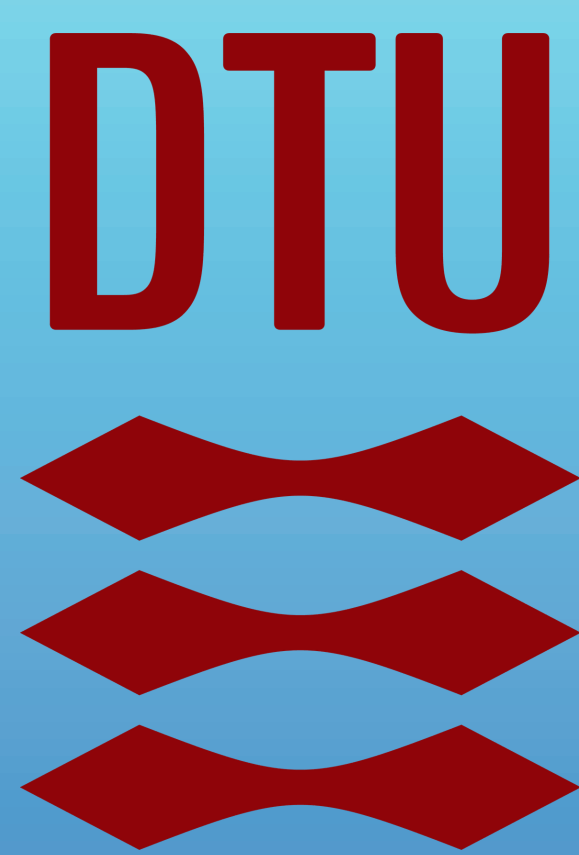


Automatic waste bin

Final Project - Group 16

34315 - Internet of things - application and infrastructure implementation



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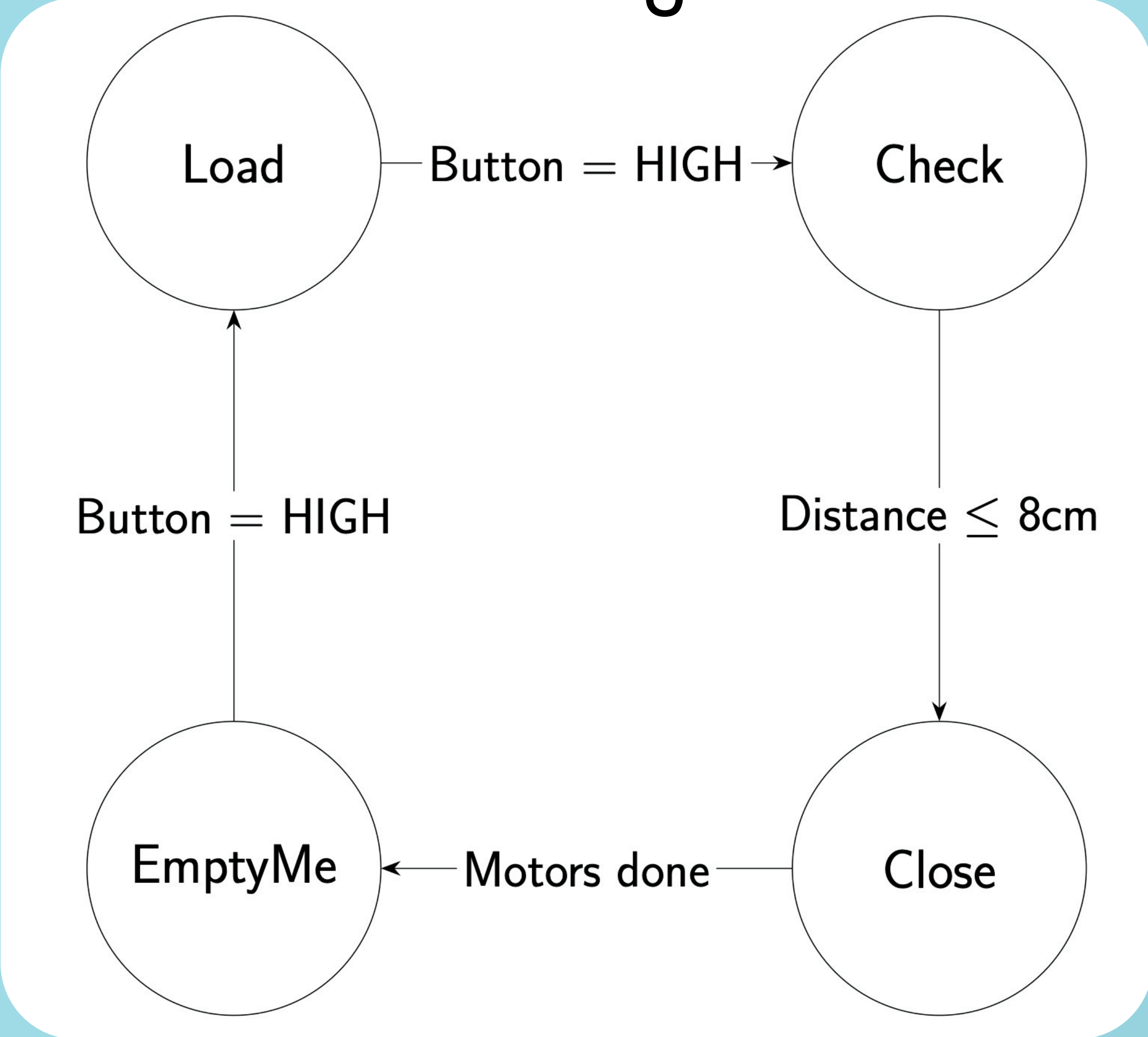
Tobias Nilsson s233987

Sigurd Hestbech s245534

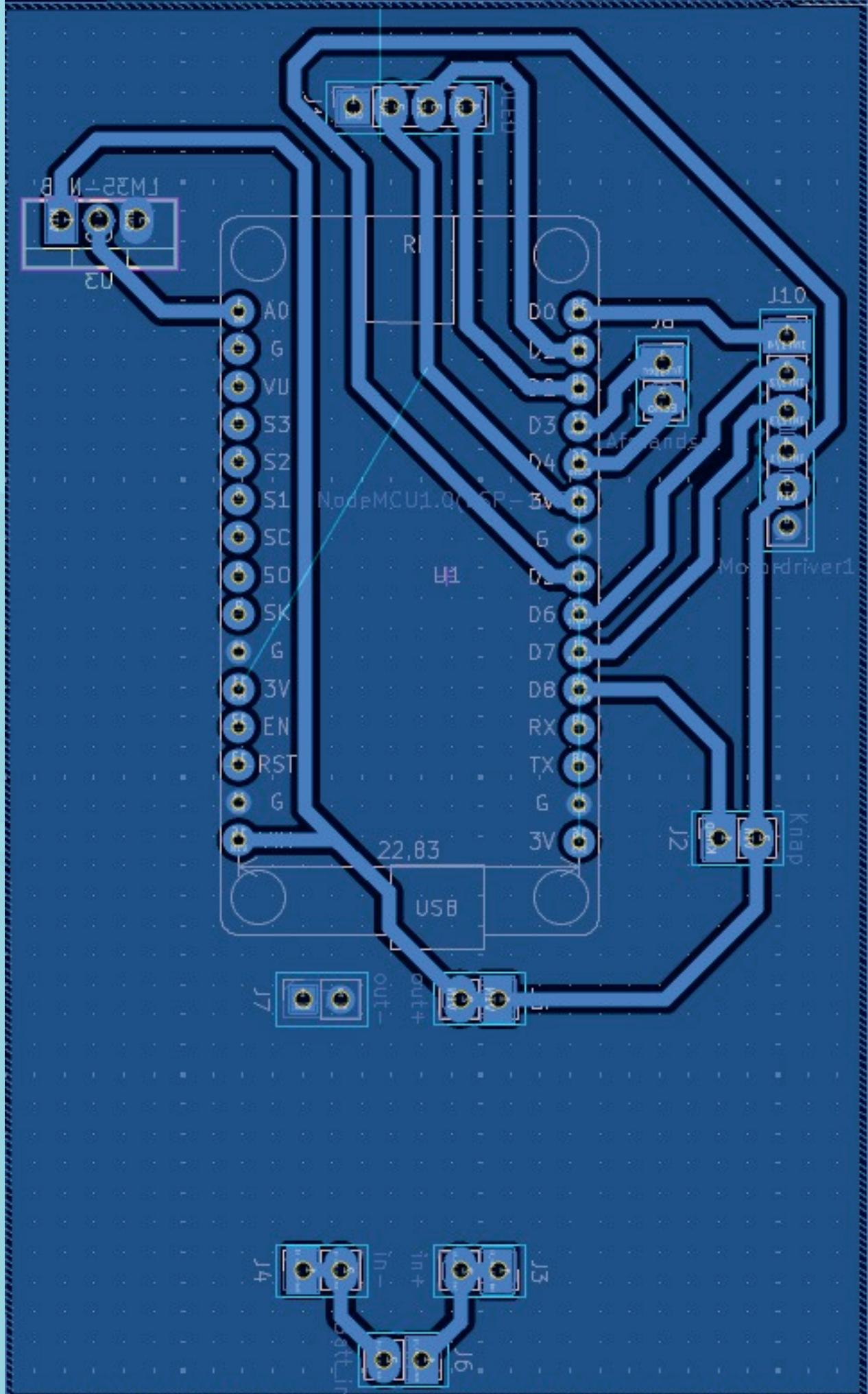
Mads Rudolph s246132

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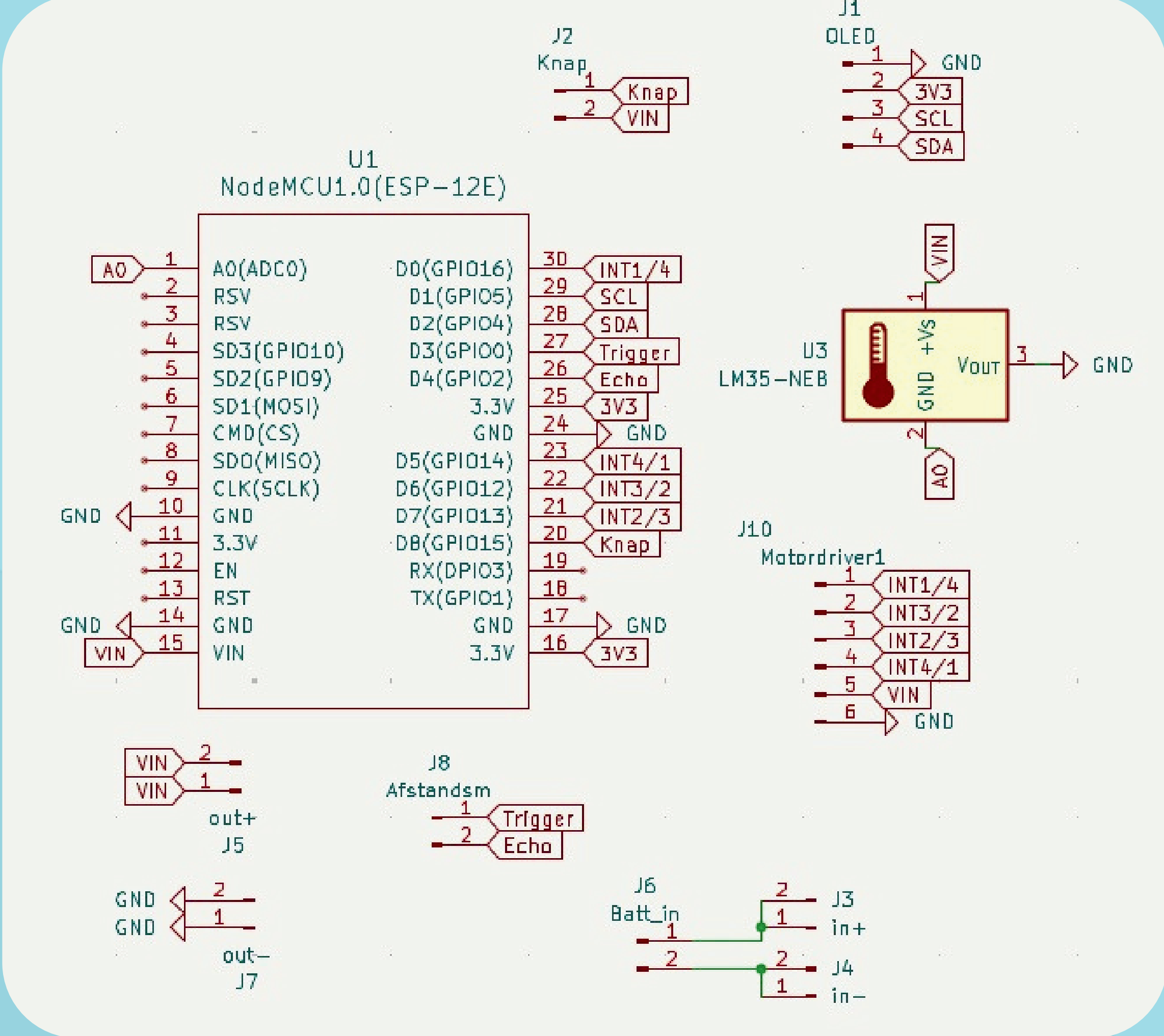
State Diagram



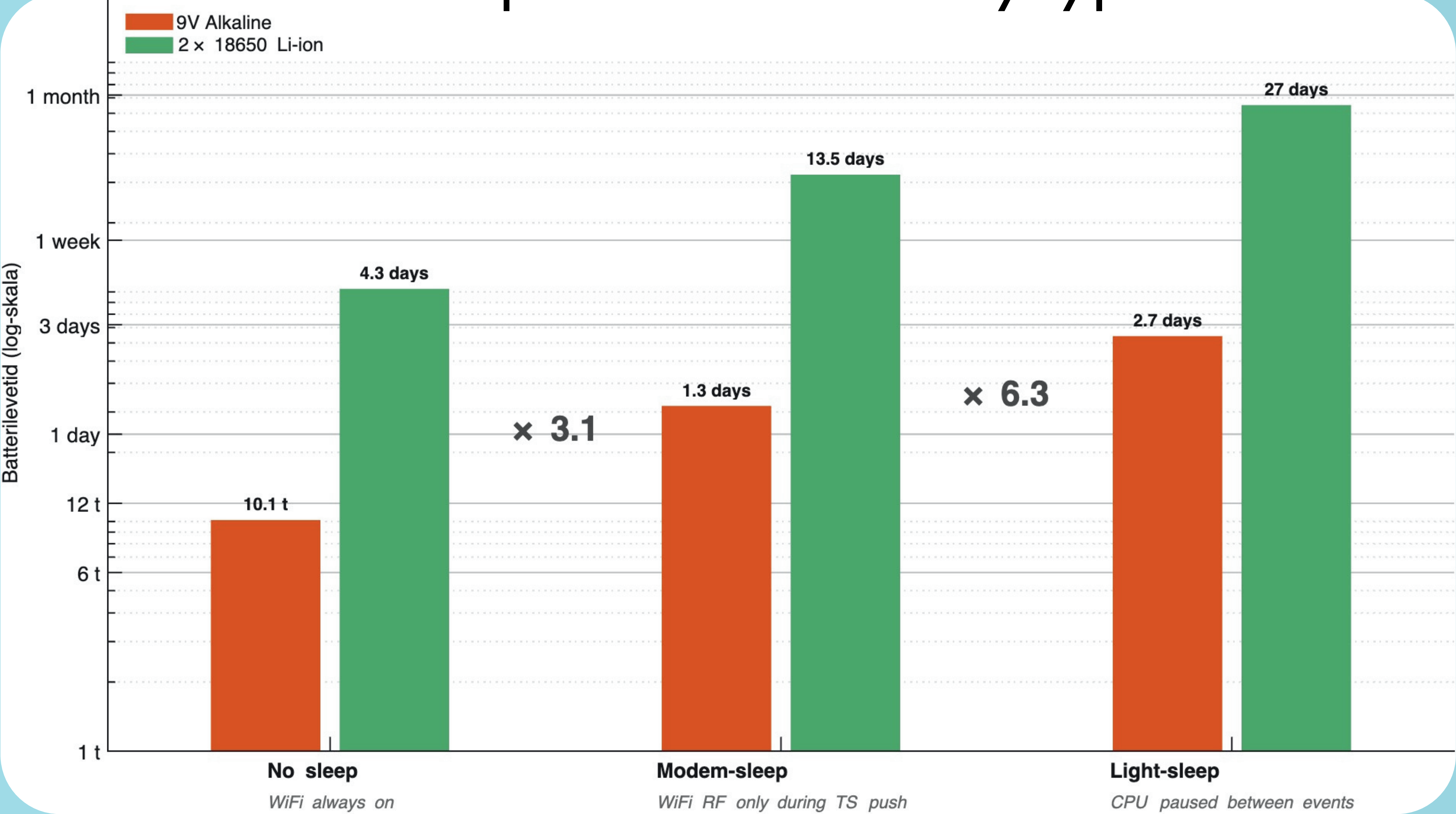
PCB



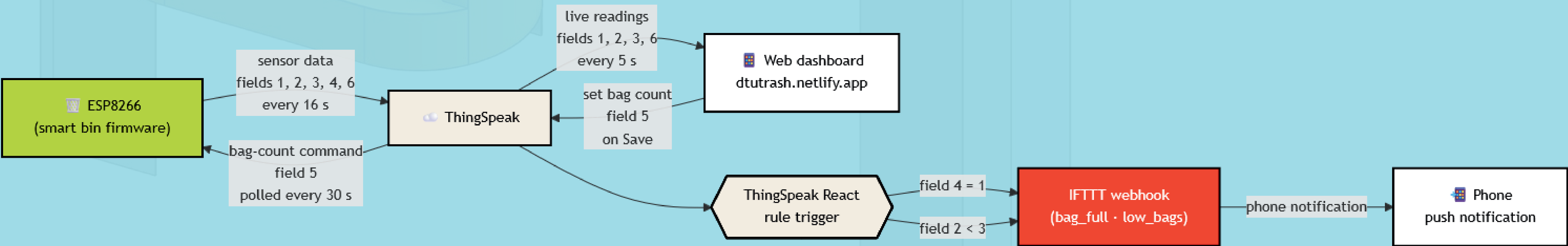
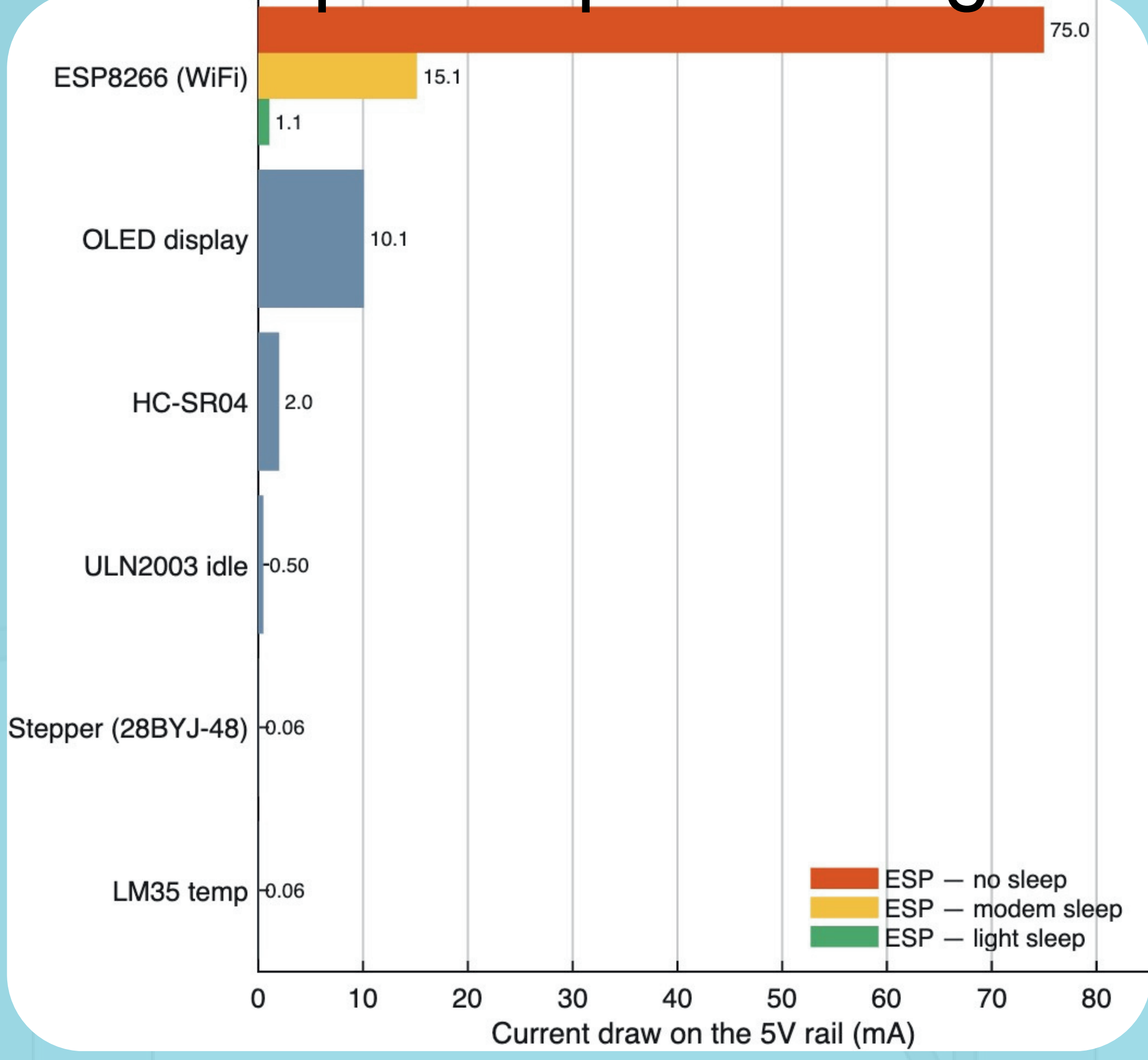
Schematic



ESP-sleep modes vs. battery type



Component power usage



ThingsPeak Fields

Field	Meaning	Written by	Read by
1	Distance (cm) from ultrasonic sensor	ESP	Dashboard
2	Bags remaining	ESP	Dashboard, IFTTT React
3	State (0=LOAD, 1=CHECK, 2=CLOSE, 3=EMPTY_ME)	ESP	Dashboard
4	Bag-full event (1 = full)	ESP	IFTTT React
5	Set-bag-count command	Dashboard (user)	ESP
6	Temperature (°C) from LM35	ESP	Dashboard

Timing

Direction	Cadence
ESP → ThingSpeak (push)	every 16 s, when data has changed
ESP ← ThingSpeak (poll field 5)	every 30 s
Dashboard → ThingSpeak (refresh)	every 5 s
Dashboard → ThingSpeak (set bag count)	on user click
ThingSpeak React → IFTTT	event-driven
IFTTT → phone	event-driven

Try Me!

